Week 1. By Deng Yufan.

Prob 1. Decide whether the following is a λ term.

a. $\lambda x.xxx$.

It is.

b. $\lambda\lambda x.x$.

It's not. There is no virable after the first λ.

c. $\lambda y.(\lambda x.x)$.

It is.

d. $\lambda uv.((xxy)xy)y)$.

It's not. Parentheses are invalid such that closed ones are more than open ones.

Prob 2. Compute the terms.

a. $(xyz)[y/z]$.

xyy.

b. $(\lambda x.x)[y/z]$.

$\lambda x.x$.

c. $(\lambda y.xy)[yy/x]$

$\lambda y'.yy\,y'$.

Prob 3. Prove equalities in $\lambda\beta$.

a. $\lambda uv.v = \lambda u.(\lambda x.x)(\lambda u.u)$.

$$\dfrac{\dfrac{\dfrac{\dfrac{}{(\lambda x.x)(\lambda u.u) = \lambda u.u}\,(\beta)}{\lambda u.(\lambda x.x)(\lambda u.u) = \lambda u.\lambda u.u}\,(abs) \qquad \dfrac{}{\lambda u.\lambda u.u = \lambda uv.v}\,(\alpha)}{\lambda u.(\lambda x.x)(\lambda u.u) = \lambda uv.v}\,(trans)}{\lambda uv.v = \lambda u.(\lambda x.x)(\lambda u.u)}\,(refl)$$

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b. $\lambda uv.(\lambda xy.y)uv = \lambda ab.b$

$$
\begin{array}{ll}
 & \dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{(\lambda xy.y)uv = (\lambda y.y)v}(\beta) \quad \dfrac{}{(\lambda y.y)v = v}(\beta)}{(\lambda xy.y)uv = v}(\text{trans})}{\lambda v.(\lambda xy.y)uv = \lambda v.v}(\text{abs})}{\lambda uv.(\lambda xy.y)uv = \lambda uv.v}(\text{abs}) \quad \dfrac{}{\lambda uv.v = \lambda ab.b}(\alpha)}{\lambda uv.(\lambda xy.y)uv = \lambda ab.b}(\text{trans})
\end{array}
$$

Prob 4.

a. Find term s that $stu = ut$ in $\lambda\beta$ for all t, u.

$s = \lambda xy.yx$.

b. Show that exist term s that $\lambda\beta \vdash st = ss$ for all t.

$s = \lambda x.y$.

Prob 5. Show that exist term G that s is fixed-point combinator iff $\lambda\beta \vdash Gs = s$.

You may assume fixed-point combinator has the form $\lambda x.s$.

We claim that $G = \lambda yx.x(yx)$. Assume fixed-point combinator has the form $\lambda x.s$.

If: $G(\lambda x.s) = \lambda x.s$

$\overset{(\text{def})}{\Rightarrow} (\lambda yx.x(yx))(\lambda x.s) = \lambda x.s$

$\cancel{\Rightarrow \lambda x.x[(\lambda x.s)x] = \lambda x.s}$. $\overset{(\beta)}{\Rightarrow} \lambda x.x[(\lambda x.s)x] = \lambda x.s$.

$\cancel{\Rightarrow \lambda x.xs = \lambda x.s}$. $\overset{(\text{app})}{\Rightarrow} [\lambda x.x[(\lambda x.s)x]]f = (\lambda x.s)f$.

$\cancel{\Rightarrow (\lambda x.xs)f = (\lambda x.s)f}$. $\overset{(\beta)}{\Rightarrow} f[(\lambda x.s)f] = (\lambda x.s)f$. for all f.

$\cancel{\Rightarrow fs[f/x] = s[f/x]}$.

$\cancel{\Rightarrow f[(\lambda x.s)f] = (\lambda x.s)f, \text{ for all } f}$.

That's what we need.

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Only If: $f[(\lambda x.s)f] = (\lambda x.s)f$. for all f.

(apply x to f).

$\Rightarrow x[(\lambda x.s)x] = (\lambda x.s)x$.

$\Rightarrow xs = s$.

$\Rightarrow \lambda x.xs = \lambda x.s$.

$\Rightarrow \lambda x.x[(\lambda x.s)x] = \lambda x.s$.

$\Rightarrow$

Only If: $f[(\lambda x.s)f] = (\lambda x.s)f$ for all f.

(apply x to f).

$\Rightarrow x[(\lambda x.s)x] = (\lambda x.s)x$.

$\overset{(\beta)}{\Rightarrow} x[(\lambda x.s)x] = s$.

$\overset{(abs)}{\Rightarrow} \lambda x.x[(\lambda x.s)x] = \lambda x.s$.

$\overset{(\beta)}{\Rightarrow} (\lambda yx.x(yx))(\lambda x.s) = \lambda x.s$.

$\overset{(def)}{\Rightarrow} G(\lambda x.s) = \lambda x.s$.

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